DocMind AI - Field Handbook

Version 2.0 | Retrieval-Augmented Generation over private documents

1. PURPOSE

This handbook is the reference document bundled with DocMind AI so that the system can be demonstrated immediately, without uploading a file. It is written to exercise every part of the retrieval pipeline: definitions, numbered settings, section headers, and facts that span two pages.

2. WHAT RETRIEVAL-AUGMENTED GENERATION IS

Retrieval-Augmented Generation is a technique that grounds a language model answer in documents fetched at question time, rather than relying on the parameters the model learned during training. The retrieved text is placed into the prompt as context, and the model is instructed to answer only from that context.

The benefit is twofold. Answers stay current, because updating the corpus requires no retraining. Answers stay auditable, because every claim can be traced to a specific page of a specific source document.

3. WHY GROUNDING MATTERS

A language model asked a question outside its knowledge will often produce a fluent but fabricated answer. This failure is called hallucination. Grounding reduces hallucination by making the correct answer available in the prompt and by giving the system a defensible way to say it does not know when retrieval returns nothing relevant.

4. THE INGESTION PIPELINE

4.1 TEXT EXTRACTION

A PDF is read page by page. Page numbers are preserved at extraction time, because a citation that cannot name a page is not verifiable. Pages that yield no text are skipped; a document that yields no text at all is rejected as scanned or empty rather than indexed as a silent failure.

4.2 CHUNKING

Extracted text is split recursively on paragraph, sentence, and word boundaries. The default chunk size is 1000 characters with an overlap of 150 characters, an overlap of fifteen percent.

Chunk size is a trade-off. Chunks that are too small lose the surrounding argument and retrieve confidently but answer incompletely. Chunks that are too large dilute the embedding, because a single vector must represent several unrelated ideas at once. Overlap exists so that a sentence falling across a boundary still appears whole in one of the two neighbours.

4.3 EMBEDDING

Each chunk is converted into a dense vector by an embedding model. Two passages with similar meaning receive vectors that sit close together, so meaning-based search becomes a nearest-neighbour lookup. Chunks are embedded in batches of up to 500 to reduce the number of API round trips.

4.4 INDEXING

Vectors are stored in a FAISS index, one index per document, written to disk. Because each document is a separate index, a document can be deleted by removing its directory, with no global rebuild.

5. THE QUERY PIPELINE

5.1 HYBRID RETRIEVAL

DocMind AI retrieves with two methods at once and merges the result.

- Dense vector search finds passages that mean the same thing as the question even when they share no vocabulary with it.
- BM25 keyword search finds passages containing the exact rare terms of the question, such as an identifier, an acronym, or a product name.

Each method fails where the other succeeds. Vector search misses exact identifiers because embeddings blur precise tokens. Keyword search misses paraphrase because it cannot match words it has never seen. Running both and combining their scores is called hybrid retrieval.

5.2 SCORE FUSION

The system fetches three times the requested number of candidates, scores each candidate by both methods, and blends them. The default weighting is 0.6 for the vector score and 0.4 for the BM25 score.

5.3 RELEVANCE THRESHOLD

A candidate whose blended score falls below 0.50 is discarded rather than returned. This is deliberate. Returning the least-bad chunk when nothing relevant exists is what causes a grounded system to hallucinate anyway. An empty result is a correct result when the corpus lacks the answer.

5.4 CONTEXT EXPANSION

Once a chunk is selected, the chunk immediately following it is appended. A retrieved passage often ends mid-argument, and the sentence that completes it lives in the next chunk.

6. THE ANSWERING STAGE

The surviving chunks are assembled into a context block and sent to the language model with an instruction to answer only from that context and to cite the page it used. The default number of chunks passed to the model is 8, controlled by the setting named TOP K.

6.1 MULTI-HOP QUESTIONS

Some questions cannot be answered by a single retrieval. A question that compares two things, or that requires one fact in order to look up a second, needs more than one pass. For these, a planner first decomposes the question into two or three focused sub-queries, each sub-query is retrieved separately, and the merged evidence is synthesised into one answer. A final verification step checks the drafted claims back against the retrieved context before the answer is shown.

7. EVALUATION

Retrieval quality is measured, not assumed. The benchmark corpus contains 1200 chunks and 60 questions with known correct answers. The metrics used are Recall at k, which asks whether the correct passage appeared at all, and Mean Reciprocal Rank, which asks how near the top it appeared.

Measuring both matters, because a change can improve one and damage the other. Adding candidates usually raises recall while lowering precision.

8. KNOWN LIMITATIONS

- Scanned PDFs without a text layer cannot be read; no OCR is performed.
- Tables lose their column structure during text extraction.
- Answers are only as good as the corpus; the system cannot know a fact that no uploaded document contains.